

Performance Testing

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Crop Prediction (/predict)
Test Environment	Cloud(Render Webservice)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Access the Home Page on Render Cloud	1	30	Home page loads successfully
2	Navigate to Find Your Crop page	1	30	Crop input form is displayed
3	Submit valid crop parameters	5	60	Recommended crop is displayed correctly
4	Submit invalid or empty input values	5	30	Validation message or error is displayed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 s	Pass	Good
2	Response Time (Max)	< 5 seconds	2.8 s	Pass	Good
3	Throughput (Req/sec)	-	18 req/sec	Pass	Stable
4	Error Rate	< 1%	0 %	Pass	No errors
5	CPU Utilization	< 80%	42 %	Pass	Normal
6	Memory Utilization	< 80%	55 %	Pass	Normal

Step 4: Observations & Analysis

Key Findings:

The application successfully generated crop recommendations with stable performance under normal load.

Bottlenecks:

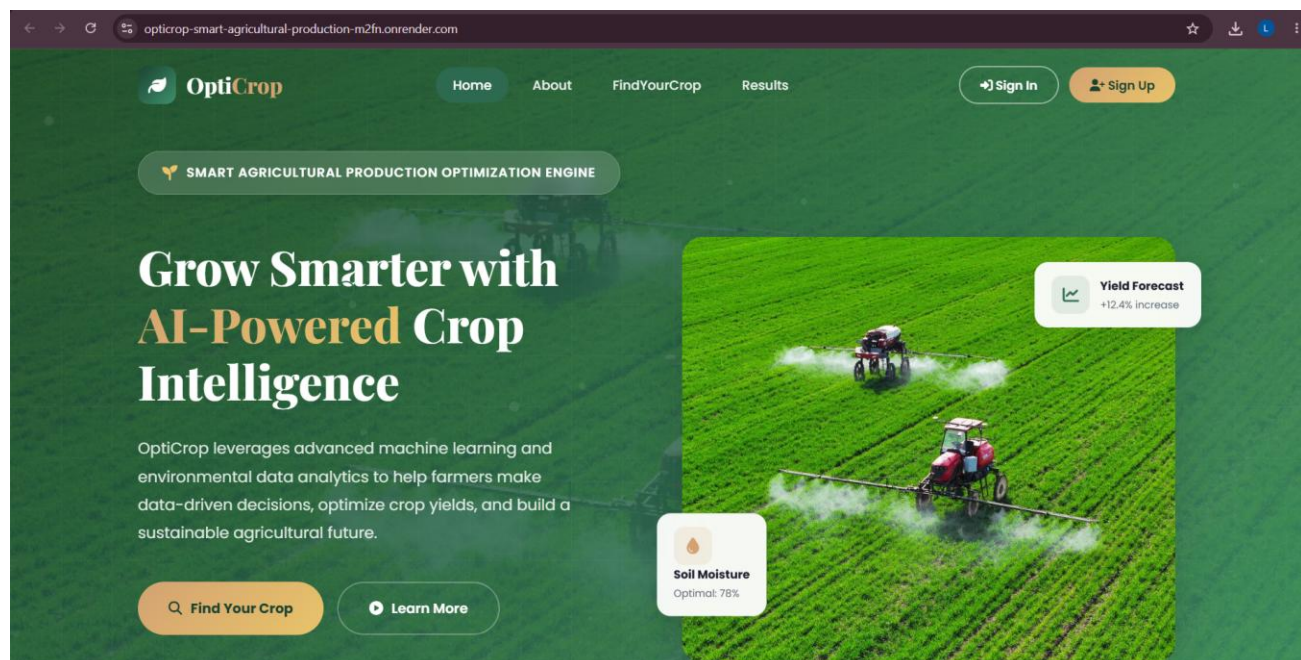
No major bottlenecks observed.

Optimization:

Model loaded once at startup and input validation improved response consistency

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



 OptiCrop

HomeAboutAdmin Panel

Hi, adminSign Out

PREDICTION REPORT

Agricultural Suitability Report

Review your personalized farm optimization analysis based on our machine learning algorithm.



BEST RECOMMENDED CROP

Coffee

AI Classifier Confidence

94%

EXPECTED YIELD

CROP SEASON

WATER NEEDS

SUNLIGHT NEEDS

Print Report

Dashboard



N

NITROGEN

80 kg/ha

P

PHOSPHORUS

30 kg/ha

K

POTASSIUM

30 kg/ha

*

SOIL PH

7.0

TEMP

27.0°C

HUMIDITY

50.0%

RAINFALL

111mm



- Requires acidic, deep volcanic or organic-rich hillside soils.
- Prune vertical suckers to channel energy into lateral branches.
- Hand-pick only the fully ripe coffee berries.
- Conduct organic mulching to preserve moisture in roots.
- Periodically perform soil health tests at local agronomy units to audit nutrient depletions.

* This recommendation is generated via an AI classifier trained on multi-attribute parameters. Follow local expert advice before implementing massive cropping cycles.

